

Cardiovascular consequences of obese and nonobese obstructive sleep apnea.

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Current evidence suggests a role for obstructive sleep apnea (OSA) in the development of cardiovascular disorders. However, obesity is an active confounder in this relationship. OSA and obesity share similar pathophysiologic mechanisms potentially leading to cardiovascular disorders. Presence of OSA in obese patients may further contribute to adverse cardiovascular outcomes when compared with each condition in isolation. In this review the authors explore the complex relationship between OSA and obesity (and nonobese subjects) in the development of cardiovascular disorders.

Postnatal Growth Trajectories and Risk of Obstructive Sleep Apnea in Middle Age: A Cohort Study.

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Rapid growth in childhood predisposes to obesity and cardiometabolic diseases in adulthood. While obstructive sleep apnea (OSA) is bidirectionally linked to obesity, its developmental origins are sparsely studied. We examined associations between postnatal growth and the risk of OSA in adulthood. We included adults whose childhood anthropometric data was collected in the New Delhi Birth Cohort study. The risk of OSA was defined by the Berlin Questionnaire (BQ) with and without the obesity criterion. Using logistic regression, we studied associations of OSA risk with conditional growth parameters, which are statistically independent measures of gain in height, weight, and body mass index (BMI), during infancy (0-2 years), early childhood (2-5 years), and late childhood (5-11 years). Among 521 subjects (58.9% males) with a mean (SD) age of 40.9 (1.7) years, 30.9% had a high risk of OSA. On multivariate analysis, a high risk of OSA was associated with a higher conditional BMI in infancy (odds ratio: 1.25; 95% confidence interval: 1.00-1.57; $p = 0.048$) and early childhood (1.35; 1.07-1.69; $p = 0.011$). Higher risk of OSA was associated with greater conditional weight in early childhood (1.34; 1.06-1.68; $p = 0.013$). Using the modified BQ definition without obesity, adult risk of OSA was significantly associated with a higher adult BMI instead of childhood conditional BMIs. Greater gain in conditional BMI or weight in early childhood is associated with a high risk of OSA in middle age, which is mediated by a higher attained adult BMI.

Current child, but not maternal, snoring is bi-directionally related to adiposity and cardiometabolic risk markers: A cross-sectional and a prospective cohort analysis.

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Obstructive sleep apnea (OSA), typically manifested as snoring, is closely associated with obesity. However, the directionality of associations of OSA with cardiometabolic risk markers is unclear, as obesity increases risk for OSA, and OSA results in excess weight gain and its metabolic consequences. Less is known about how obesity and OSA may relate in children and adolescents and whether maternal OSA may influence the development of obesity and cardiometabolic dysfunction in offspring. Among 1078 children from the Project Viva cohort, we examined cross-sectionally and prospectively associations of parent-reported child or maternal snoring with cardiometabolic outcomes, including adiposity, adipokines, and insulin resistance. Cross-sectionally, child snoring was related to adiposity and metabolic

risk, particularly body mass index (BMI; β 0.61kg/m², 95% CI 0.33, 0.89; $p < 0.001$), trunk fat mass index (β 0.23kg/m², CI 0.12, 0.34; $p < 0.001$), high-density lipoprotein cholesterol (β -1.47mg/dL, CI -2.69, -0.25; $p = 0.02$), and metabolic risk z-score (β 0.08, CI 0.02, 0.14; $p = 0.01$) after correction for covariates. Prospectively, adiposity (BMI, trunk fat, fat mass, and waist circumference) and cardiometabolic (leptin, HOMA-IR, CRP, and global metabolic risk) measures at mid-childhood (~7y) were associated with child snoring at the early teen visit (~12y) after correction for covariates. Child snoring at ~9y was related to changes in adiposity between mid-childhood and early teen visits. Child but not maternal snoring, was related to child adiposity and cardiometabolic outcomes. Adiposity and child snoring are associated with each other cross-sectionally and are each predictive of the other among children/adolescents prospectively. These results suggest similar mechanisms in pediatric/adolescent populations as in adults for the development of sleep-disordered breathing and sleep apnea that will need to be confirmed in randomized clinical trials. Importantly, this research points to the need to target both sleep and obesity in order to break this vicious cycle.

A pilot study on the biomechanical assessment of obstructive sleep apnea pre and post bariatric surgery.

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Obesity is a major risk factor for obstructive sleep apnea patients. In obese patients the severity of this risk can be reduced by bariatric surgery. This pilot study investigates the perioperative effects of bariatric surgery on obstructive sleep apnea and on the physical and biomechanical characteristics of the upper airway. Polysomnography and computer tomography data for 10 morbid obese patients promoted for bariatric surgery were conducted before surgery and at 6 and 12 months postoperatively for assessment of the oropharyngeal anatomy, and subsequent three-dimensional modelling of the airway. Mean values for the apnea/hypopnea index and body mass index significantly reduced after surgery. To combine the effect of changes in the upper airway volume and body mass index, a new volume body mass index is introduced. This index increases with a successful bariatric surgery. Although bariatric surgery leads to an effective weight reduction for all age groups, for obstructive sleep apnea patients it may be effective for middle age, less effective for 50-60 years, and further less effective for patients over the age of 60 years.

Dynamic Upper Airway Imaging during Wakefulness in Obese Subjects with and without Sleep Apnea.

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Obesity is a major risk factor for obstructive sleep apnea. Although greater dimensional changes in the upper airway during wake respiration have been noted in patients with apnea compared with control subjects, whether these differences remain in the presence of obesity is unknown. To evaluate upper airway anatomic characteristics and airway compliance (distensibility) in obese subjects with obstructive sleep apnea compared with obese control subjects. Dynamic magnetic resonance imaging was performed in 157 obese subjects with apnea and 46 obese control subjects during wakefulness in the midsagittal and three axial upper airway regions (retropalatal, retroglossal, epiglottal). Differences in measurements between subjects with apnea and control subjects, and correlations with apnea-hypopnea index among

subjects with apnea, were examined. Measurements included airway areas and linear dimensions. Subject-specific coefficients of variation were calculated to examine variability in airway size. Controlling for covariates, the retropalatal area during respiration was significantly smaller in subjects with apnea than control subjects, based on the average ($P = 0.003$), maximum ($P = 0.004$), and minimum ($P = 0.001$) airway area. Airway narrowing was observed in anteroposterior and lateral dimensions (adjusted $P < 0.05$). Results were similar in an age, sex, and body mass index-matched subsample. There were significant correlations between apnea-hypopnea index and dynamic measures of airway caliber in the retropalatal and retroglossal regions among subjects with apnea. Upper airway caliber during respiration was significantly narrower in obese subjects with apnea than obese control subjects in the retropalatal region. These findings provide further evidence that retropalatal airway narrowing plays an important role in the pathogenesis of obstructive sleep apnea in obese subjects.

Sleep apnea and hypertension.

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Obstructive sleep apnea (OSA) is the most common form of sleep-disordered breathing and frequently coexists with obesity. Almost 15 million Americans are affected by this disorder. This prevalence is likely increasing, given the current epidemic of obesity. Recent data confirm an association between sleep apnea and several cardiovascular disease conditions, suggesting that OSA may be a new risk factor for coronary artery disease, heart failure, heart rhythm disturbances and hypertension, independent of body mass index. In this review, the authors focus on the nature of the association between OSA and hypertension, the evidence suggesting a causal interaction, and discuss the potential pathophysiologic mechanisms responsible. These mechanisms include activation of the sympathetic and renin-angiotensin-aldosterone systems (RAAS), oxidative stress, and systemic and vascular inflammation, all of which could link OSA to a sustained increase in blood pressure. The authors also review potential therapeutic strategies for the hypertensive patient with OSA.

Utility of Screening Questionnaires to Detect Obstructive Sleep Apnea in Patients with Obesity.

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Obesity is one of today's most concerning health problems due to increased cardiovascular risk, which is still the leading cause of death. Obstructive sleep apnea syndrome (OSAS) is certainly one of the important risk factors that links obesity and cardiovascular risk. There is a great need to evaluate obstructive sleep apnea (OSA) in obese patients. Today, there are easily available and applicable questionnaires (Epworth Sleepiness Scale (ESS), STOP, STOP-Bang (SBQ), Insomnia Severity Index (ISI) and Pittsburgh Sleep Quality Index (PSQI)) that could be very useful in clinical practice for this very purpose. The aim of this paper is to investigate the sensitivity and specificity of the questionnaires for OSA screening in obese patients with and without OSA. This cross-sectional study was carried out in the tertiary healthcare centre. The following questionnaires were used: ESS, STOP, SBQ, ISI, PSQI. 70 (58 female) adult patients with obesity (body mass index (BMI) $> 30 \text{ kg/m}^2$) were included. SBQ showed sensitivity of 75%, specificity of 75% at cut-off of 5.5 with the Youden index of 0.5, while PSQI had sensitivity of 78%, specificity of 67% at cut-off of 17.75 with slightly smaller Youden index 0.45. STOP and

ESS had a sensitivity of 77% and 75%, respectively but with an even smaller Youden index (0.23 and 0.21), and ISI had the lowest sensitivity of 59% and the lowest Youden index (0.13) of the questionnaires we examined. Our study results suggest that SBQ and PSQI are best screening tools in detecting OSA in patients with obesity. Further study of these questionnaires and possible modifications are certainly important for future research.

beta3-Adrenergic receptor Trp64Arg polymorphism and increased body mass index in sleep apnoea.

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Obesity is an important risk factor for obstructive sleep apnoea syndrome (OSAS), insulin resistance and cardiovascular disease. The substitution of tryptophan 64 with arginine (Trp64Arg) polymorphism (Arg variant) of the beta(3)-adrenergic receptor (ADRB3) has been associated with obesity. In this study, the prevalence of the Trp64Arg ADRB3 polymorphism in a large group of patients with OSAS and its association with body mass index (BMI), insulin resistance and hypertension were evaluated. ADRB3 genotype was determined in 387 patients with OSAS and 137 healthy subjects recruited from three Spanish tertiary hospitals. The distributions of the ADRB3 genotypes were similar in OSAS and controls, and, in a multivariate model, the risk of OSAS was not associated with the presence of the Arg variant of the ADRB3 gene. However, BMI was higher in those patients with OSAS who carried this genetic variant than in those with the Trp variant. Furthermore, a linear trend for higher BMI was found in those with the Arg variant (56, 75 and 100% for Trp/Trp, Trp/Arg and Arg/Arg, respectively). Insulin resistance, blood pressures and serum levels of lipids and glucose were not associated with the presence of the Arg variant of the ADRB3 gene. The presence of the arginine 64 allele of the beta(3)-adrenergic receptor gene does not increase the risk of obstructive sleep apnoea syndrome, but is associated with the development of obesity in those patients who suffer obstructive sleep apnoea syndrome.

The impact of obesity in cognitive and memory dysfunction in obstructive sleep apnea syndrome.

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Obstructive sleep apnea (OSA), a sleep disorder, results in decreased daytime alertness and neurocognitive dysfunction. Obesity is considered a major risk factor for the development and progression of OSA and the resulting cognitive dysfunction. However, the effect of obesity on neurocognitive dysfunction in OSA has been rarely investigated. Eighty-three patients with moderate to severe OSA syndrome were recruited in our study. After matching for education, age, and body mass index (BMI), 40 patients were enrolled into our study with matched obese (BMI ≥ 30) and non-obese (BMI < 30) groups. All enrolled patients completed a polysomnographic study, sleepiness questionnaires, and attention, cognitive, and memory function tests. Compared to obese OSA patients, non-obese OSA patients had shorter reaction times in the psychomotor vigilance task but not the Flanker or Stroop cognitive tasks. Additionally, obese OSA patients had a reduced capacity for working memory relative to non-obese OSA patients. Obesity had a significant effect on OSA patients in our study, including delayed reaction times in the psychomotor vigilance task and a decrease in working memory.

Obstructive sleep apnea and cardiovascular disease.

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Obstructive sleep apnea (OSA) is a common disorder associated with an increased risk of cardiovascular disease and stroke. As it is strongly associated with known cardiovascular risk factors, including obesity, insulin resistance, and dyslipidemia, OSA is an independent risk factor for hypertension and has also been implicated in the pathogenesis of congestive cardiac failure, pulmonary hypertension, arrhythmias, and atherosclerosis. Obesity is strongly linked to an increased risk of OSA, and weight loss can reduce the severity of OSA. The current standard treatment for OSA-nasal continuous positive airway pressure (CPAP)-eliminates apnea and the ensuing acute hemodynamic changes during sleep. Long-term CPAP treatment studies have shown a reduction in nocturnal cardiac ischemic episodes and improvements in daytime blood pressure levels and left ventricular function. Despite the availability of effective therapy, OSA remains an underdiagnosed and undertreated condition. A lack of physician awareness is one of the primary reasons for this deficit in diagnosis and treatment.
